



HILL CIPHER TECHNIQUE

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INTRODUCTION

- The **Hill Cipher** is a **polygraphic substitution cipher** developed by **Lester S. Hill** in **1929**.
- It uses **matrix multiplication** and **modular arithmetic** to encrypt groups of letters instead of individual characters.
- Unlike simple substitution ciphers, the Hill Cipher encrypts multiple letters at a time, making it more resistant to frequency analysis.

- **Aim:**

The **Hill Cipher** is to securely encrypt plaintext by using **matrix multiplication** and **modular arithmetic**. It converts blocks of letters into ciphertext, providing better security than simple substitution ciphers and demonstrating the application of linear algebra in cryptography.

Algorithm

Encryption Algorithm

- Choose an invertible key matrix **K**.
- Convert plaintext letters into numbers (A = 0, B = 1, ..., Z = 25).
- Divide the plaintext into equal-sized blocks.
- Multiply each plaintext block by the key matrix.
- Perform modulo 26 on the result.
- Convert the resulting numbers back into letters to obtain the ciphertext.

Decryption

- Find the inverse of the key matrix modulo 26.
- Multiply each ciphertext block by the inverse key matrix.
- Apply modulo 26.
- Convert the numbers back into letters to recover the plaintext.

EXAMPLE

HILL CIPHER – EXAMPLE

1. KEY MATRIX

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

2. PLAINTEXT

Plaintext : **HI**

Convert letters to numbers (A = 0, B = 1, ..., Z = 25)

$$H = 7 \quad I = 8$$

$$\text{Plaintext vector (P)} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Note: All operations are performed
modulo 26

3. ENCRYPTION ($C = K \times P \bmod 26$)

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Apply modulo 26:

$$\begin{array}{l} 45 \bmod 26 \\ 54 \bmod 26 \end{array} = \begin{array}{l} 19 \\ 2 \end{array}$$

4. CONVERT NUMBERS BACK TO LETTERS

$$\begin{array}{l} 19 \rightarrow \mathbf{T} \\ 2 \rightarrow \mathbf{C} \end{array}$$

CIPHERTEXT : TC

ADVANTAGES

- Encrypts multiple letters simultaneously.
- More secure than simple substitution ciphers.
- Resistant to frequency analysis.
- Uses mathematical concepts for stronger encryption.
- Fast encryption for block data.

DISADVANTAGES

- Requires an invertible key matrix. Key generation is more complex.
- Difficult to perform manually.
- Small block sizes reduce security.
- Vulnerable if the key matrix is discovered.